Supervyse OPF Default User Configuration Guide

Revision 1.2

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Revision History

Revision Number	Date	Author	Reviewer	Summary
1.0	2024-10-24	Roy Kuo	Brad Chang	Initial version
1.1	2024-12-12	Roy Kuo	Brad Chang	Add section "Error handling"
1.2	2025-01-22	Roy Kuo	Brad Chang	Release version

1 Default User Configuration Guide

1.1 Introduction

This page describes how to configure default username and password in OPF firmware.

- Configure default username and password at build time
  - It means all platforms use the same default user with this firmware
- Store default username and password in EEPROM
  - It means default user can be difference in each platform with this firmware
  - IPMI OEM command to set default user / password in EEPROM
- The flag for **CA law SB-327**:
  - Force the default user to update password when the default user login at first time.

1.2 Configure default username and password at build time

1.2.1 Configuration

The configuration file path depends on vendor and target platform:

- The configuration file of CRBs in **OPF G3** SDK:
  - For Intel **ArcherCity** CRB

```
$(OPF_SDK)/meta-insyde/meta-intel/meta-archercity/conf/local.conf.sample
```

  - For AMD **Onyx** / **Quartz** CRBs

```
$(OPF_SDK)/meta-insyde/meta-amd/meta-sp5/conf/local.conf.sample
```

  - For AMD **Turin** CRB

```
$(OPF_SDK)/meta-insyde/meta-amd/meta-turin/conf/local.conf.sample
```

  - For Nvidia **Falcon** / **Starship** / **MGX CG1** / **MGX C2** CRBs

```
$(OPF_SDK)/meta-insyde/meta-nvidia/meta-gh/conf/local.conf.sample
```

  - For **ast2600evb**

```
$(OPF_SDK)/meta-insyde/meta-evb/meta-evb-ast2600/conf/local.conf.sample
```

- The configuration file of CRBs in **OPF G4** SDK:
  - For Intel **BeechnutCity** / **AvenueCity** CRBs

```
$(OPF_SDK)/meta-insyde/meta-intel/meta-bhs/conf/templates/default/local.conf.sample
$(OPF_SDK)/meta-insyde/meta-intel/meta-bhs/conf/templates/advanced/local.conf.sample
```

  - For Nvidia **Oberon** CRB

```
$(OPF_SDK)/meta-insyde/meta-nvidia/meta-gb200nvl/conf/templates/default/local.conf.sample
```

  - For **ast2600evb**

```
$(OPF_SDK)/meta-insyde/meta-evb/meta-ast2600evb/conf/templates/default/local.conf.sample
```

  - For **ast2700evb**

```
$(OPF_SDK)/meta-insyde/meta-evb/meta-ast2700evb/conf/templates/default/local.conf.sample
```

- The configuration file of **new platform** created by customer:

```
$(OPF_SDK)/meta-base/meta-${new_platform}/conf/templates/default/local.conf.sample
```

Below is an example with following settings:

- username is **test**
- password is **0penBmc^**

Configuration in file:

```
INSYDE_DEFAULT_USER = "test"
INSYDE_DEFAULT_PASSWD = "0penBmc^"
```

1.2.2 Limitation

- **Only one** account can be configured, and this account has **root** permissions.
- Username / password must conform IPMI specification.
  - Length of username must equal or less than **16** bytes
  - Length of password must equal or less than **20** bytes
- Username must match regular expression "**^[a-zA-Z][a-zA-Z0-9_]***"
- Password can't use the characters "****" and "**$**", it will make python parsing error.

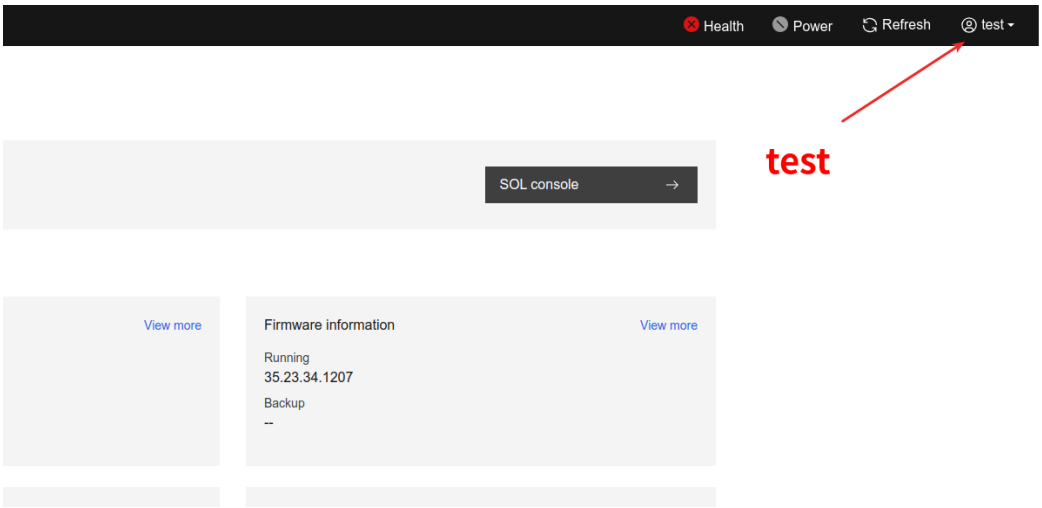
1.2.3 Verification

Test default username and password by login below interfaces

Test login BMC debug console (UART)

```
Phosphor OpenBMC (Phosphor OpenBMC Project Reference Distro) nodistro.0 intel-obmc ttyS4
intel-obmc login: test
Password:
test@intel-obmc:~#
```

Test login WebUI



Test login with SSH

```
ssh test@192.168.88.230
test@192.168.88.230's password:
test@intel-obmc:~#
```

Test IPMI command

```
ipmitool -U test -P 0penBmc^ -I lanplus -C 17 -H 192.168.88.230 raw 6 1
 22 80 23 23 02 bf 09 b0 00 00 01 00 07 12 34
```

1.3 Store default username and password in EEPROM

Store default username and **encrypted** password in EEPROM and it will be set up when BMC boot with match one of conditions below:

1. User request **Factory Reset** before reboot
2. User use IPMI OEM command **set default username and password** before reboot

1.3.1 Configuration

The configuration file **def_user.cfg** is depends on platform, only **OPF G3 Nvidia platform** has configuration currently.

```
$(OPF_SDK)/meta-insyde/meta-nvidia/meta-gh/recipes-phosphor/user/insyde-default-user-tool/def_user.cfg
```

OPF uses common settings if there are no settings for the target platform

For **OPF G3** and **OPF G4** SDK:

```
$(OPF_SDK)/meta-insyde/meta-common/meta-core/recipes-phosphor/users/insyde-default-user-tool/def_user.cfg
```

Configuration field description:

- **INSYDE_DEFAULT_USER_FROM_EEPROM**: Enable this feature if set as y
- **INSYDE_DEFAULT_USER_I2C_BUS**: EEPROM I2C bus, equal to I2C device number in filesystem (/dev/i2c-*)
- **INSYDE_DEFAULT_USER_I2C_ADDR**: EEPROM I2C address (0x00 ~ 0x7f)
- **INSYDE_DEFAULT_USER_OFFSET:** default username and password start offset in EEPROM

Below is an example with following settings:

- Enable this feature
- EEPROM I2C bus is 4
- EEPROM I2C address is 0x50
- Default username and password start offset is 0xB0

Configuration in file:

```
INSYDE_DEFAULT_USER_FROM_EEPROM=y
INSYDE_DEFAULT_USER_I2C_BUS=4
INSYDE_DEFAULT_USER_I2C_ADDR=0x50
INSYDE_DEFAULT_USER_OFFSET=0xB0
```

1.3.2 Limitation

- **Only one** account can be configured, and this account has **root** permissions.
- Username / password must conform IPMI specification.
  - Length of username must equal or less than **16** bytes
  - Length of password must equal or less than **20** bytes
- Username must match regular expression "**^[a-zA-Z][a-zA-Z0-9_]***"
- Need **80** bytes in EEPROM for store username and encrypted password

1.3.3 IPMI OEM command for set default username and password

This command will set username and password into EEPROM.

	byte	data field
Request Data	1	NetFn = Insyde OEM (3Eh)
	2	Command ID = C0h
	3	Insyde NetFn = Application (06h)
	4	Insyde Sub-Command ID = 10h
	5:20	Default username string in ASCII, Must conform REGEX="^[a-zA-Z][a-zA-Z0-9_]*", fill unset data with 0xff To clean data in EEPROM if all data in Username & Password set as 0xff
	21:40	Default password string in ASCII, fill unset data with 0xff To clean data in EEPROM if all data in Username & Password set as 0xff
Response Data	1	Completion Code

1.3.4 Error handling

If default username or encrypted password in EEPROM is **invaild**, BMC will use the default username settings which configure at build time(see detail in this document Section 1.2). If the there is no data configure at build time, use default username as OpenBMC settings. i.e. Username is **root** and password is **0penBmc**

1.3.5 Verification

Test IPMI OEM command to set data into EEPROM

Login BMC debug console (UART) and issue command, set default username as **admin** and default password as **OpenBmc**

```
ipmitool raw 0x3e 0xc0 0x06 0x10 \
 0x61 0x64 0x6d 0x69 0x6e 0xff 0xff 0xff 0xff 0xff 0xff 0xff 0xff 0xff 0xff 0xff \
 0x30 0x70 0x65 0x6E 0x42 0x6D 0x63 0xff 0xff 0xff 0xff 0xff 0xff 0xff 0xff 0xff \
 0xff 0xff 0xff 0xff
```

Use i2c tool check data in EEPROM

Below is an example:

- **i2cdump** command read data from I2C bus 7 address 0x80
- default username is **admin**
- Encrypted password data is **U2FsdGVkX19CsPOxKQ/K6qhGnJbTdNmk/sC64h4WrAs=**

```
root@bmc-maca4bf018d03a8:~# ## admin / OpenBmc
root@bmc-maca4bf018d03a8:~# ipmitool raw 0x3e 0xc0 0x06 0x10 \
> 0x61 0x64 0x6d 0x69 0x6e 0xff 0xff 0xff 0xff 0xff 0xff 0xff 0xff 0xff 0xff 0xff \
> 0x30 0x70 0x65 0x6E 0x42 0x6D 0x63 0xff 0xff 0xff 0xff 0xff 0xff 0xff 0xff 0xff 0xff 0xff 0xff 0xff

root@bmc-maca4bf018d03a8:~# i2cdump -f -y 7 80
No size specified (using byte-data access)
     0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f    0123456789abcdef
00: 01 00 00 00 01 0d 00 f1 01 0c 19 e1 53 4f 4c 55    ?...??.?????SOLU
10: 4d 20 43 4f 2e 2c 20 4c 54 44 2e 20 20 20 20 20    M CO., LTD.
20: 20 20 20 20 20 20 20 20 20 20 20 20 c8 49 53                ?IS
30: 31 36 32 46 32 32 ca 47 33 36 32 33 34 2d 30 31    162F22?G36234-01
40: 38 c3 31 33 41 dc 50 53 53 46 31 36 32 32 30 32    8?13A?PSSF162202
50: 41 43 4e 53 31 36 32 32 41 34 42 4d 41 51 32 30    ACNS1622A4BMAQ20
60: 36 30 00 00 c1 00 00 af 00 02 18 43 a3 40 06 34    60..?..?.??C?@?4
70: 08 41 02 28 23 b0 36 50 46 20 67 2f 3f 0c 0e 34    ?A?(#?6PF g/???4
80: a8 00 40 06 00 01 02 0d e6 0a 01 b0 04 74 04 ec    ?.@?.????????t??
90: 04 78 00 00 00 85 00 01 82 0d 31 3f 82 b0 04 74    ?x...?.???1????t
a0: 04 ec 04 78 00 00 00 ac 0d ff ff ff ff ff ff ff    ???x...??.......
b0: 61 64 6d 69 6e ff ff ff ff ff ff ff ff ff ff ff    admin...........
c0: 55 32 46 73 64 47 56 6b 58 31 39 43 73 50 4f 78    U2FsdGVkX19CsPOx
d0: 4b 51 2f 4b 36 71 68 47 6e 4a 62 54 64 4e 6d 6b    KQ/K6qhGnJbTdNmk
e0: 2f 73 43 36 38 68 34 57 72 41 73 3d ff ff ff ff    /sC68h4WrAs=....
f0: ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff    ................
```

Below is an example to use factory reset command (NetFn=0x30, Cmd=0x02), and reboot can see those logs:

Reboot for reload ussrname-password setting

factory reset requested

default username/password has setup

```
admin@nvidia-obmc:~# ipmitool raw 0x30 0x02 0x43 0x4c 0x52 0xcc
 00
admin@nvidia-obmc:~# reboot
Reboot for reload username-password setting
admin@nvidia-obmc:~# [  OK  ] Removed slice Slice /system/getty.
[  OK  ] Removed slice Slice /system/modprobe.
[  OK  ] Removed slice Slice /system/phosphor-reset-host-running.
[  OK  ] Removed slice Slice /system/systemd-coredump.
[  OK  ] Removed slice Slice /system/xyz.openbmc_project.Hwmon.
[  OK  ] Stopped target Bmc network online target.
[  OK  ] Stopped target Host and Network Name Lookups.
[  OK  ] Stopped target Host0 running after reset.
[  OK  ] Stopped target Timer Units.
```

```
         Stopping Phosphor Button Handler...
         Stopping Phosphor certificate manager for KEK...
         Stopping Phosphor certificate manager for KEKDefault...
         Stopping Phosphor certificate manager for PK...
         Stopping Phosphor certificate manager for PKDefault...
[ Note ] Reboot for reload username-password setting
         Stopping Phosphor certificate manager for authority...
         Stopping Phosphor certificate manager for bmcweb...
         Stopping Phosphor certificate manager for db...
         Stopping Phosphor certificate manager for dbDefault...
         Stopping Phosphor certificate manager for dbt...
         Stopping Phosphor certificate manager for dbtDefault...
         Stopping Phosphor certificate manager for dbx...
         Stopping Phosphor certificate manager for dbxDefault...
```

```
[    3.354931] mmcblk0boot1: mmc0:0001 DG4016 4.00 MiB
[    3.361930] mmcblk0rpmb: mmc0:0001 DG4016 4.00 MiB, chardev (243:0)
rofs = mtd6 squashfs rwfs = mtd10 jffs2
factory-reset
Factory reset requested.
No files will be selected for save.
update: --no-restore-files --no-save-files
The string starts with mtd
Skipping empty update of rwfs.
Clearing read-write overlay filesystem.
Erasing 64 Kibyte @ 1000000 - 100% complete.
Clearing log partition
Erasing 64 Kibyte @ 2fc0000 - 100% complete.
Restoring saved files to read-write overlay filesystem.
update: --no-save-files --clean-saved-files
The string starts with mtd
Skipping empty update of rwfs.
copy-base-filesystem-to-ram
[  134.364516] loop0: detected capacity change from 0 to 110592
[  134.435944] jffs2: notice: (253) jffs2_build_xattr_subsystem: complete building xa
[  134.496296] overlayfs: upper fs does not support tmpfile.
[  134.508081] overlayfs: upper fs does not support RENAME_WHITEOUT.
[  135.738408] jffs2: notice: (261) jffs2_build_xattr_subsystem: complete building xa
Mounting of logging partition completed
[  140.590766] random: crng init done
default username/password has set up
[  141.724637] systemd[1]: Failed to find module 'autofs4'
[  141.921485] systemd[1]: systemd 249 running in system mode (+PAM -AUDIT -SELINUX -
DN2 -IDN -IPTC +KMOD -LIBCRYPTSETUP +LIBFDISK -PCRE2 -PWQUALITY -P11KIT -QRENCODE -BZ
[  141.961668] systemd[1]: Detected architecture arm.

Welcome to NVIDIA Grace Hopper (OpenBMC     Project Reference Distro) 35.24.08.3210!
```

factory reset

default username/passwrod set up

1.4 Configure force update password at first login

CA law SB-327 is a law of the State of California which requests user to update password at first login. OPF can configure the feature which forces the user to update password at first login.

1.4.1 Configuration

The configuration file path depends on vendor and target platform:

- The configuration file of CRBs in **OPF G3** SDK:
  - For Intel **ArcherCity** CRB

```
$(OPF_SDK)/meta-insyde/meta-intel/meta-archercity/conf/local.conf.sample
```

  - For AMD **Onyx** / **Quartz** CRBs

```
$(OPF_SDK)/meta-insyde/meta-amd/meta-sp5/conf/local.conf.sample
```

  - For AMD **Turin** CRB

```
$(OPF_SDK)/meta-insyde/meta-amd/meta-turin/conf/local.conf.sample
```

  - For Nvidia **Falcon** / **Starship** / **MGX CG1** / **MGX C2** CRBs

```
$(OPF_SDK)/meta-insyde/meta-nvidia/meta-gh/conf/local.conf.sample
```

  - For **ast2600evb**

```
$(OPF_SDK)/meta-insyde/meta-evb/meta-evb-ast2600/conf/local.conf.sample
```

- The configuration file of CRBs in **OPF G4** SDK:
  - For Intel **BeechnutCity** / **AvenueCity** CRBs

```
$(OPF_SDK)/meta-insyde/meta-intel/meta-bhs/conf/templates/default/local.conf.sample
$(OPF_SDK)/meta-insyde/meta-intel/meta-bhs/conf/templates/advanced/local.conf.sample
```

  - For Nvidia **Oberon** CRB

```
$(OPF_SDK)/meta-insyde/meta-nvidia/meta-gb200nvl/conf/templates/default/local.conf.sample
```

  - For **ast2600evb**

```
$(OPF_SDK)/meta-insyde/meta-evb/meta-ast2600evb/conf/templates/default/local.conf.sample
```

  - For **ast2700evb**

```
$(OPF_SDK)/meta-insyde/meta-evb/meta-ast2700evb/conf/templates/default/local.conf.sample
```

- The configuration file of **new platform** created by customer:

```
$(OPF_SDK)/meta-base/meta-${new_platform}/conf/templates/default/local.conf.sample
```

Below is an example which enable the feature:

```
INSYDE_FIRST_LOGIN_UPDATE_PASSWD = "y"
```

1.4.2 Limitation

Only **default account** can be configured to force update password at first login

1.4.3 Verification

Enable the feature and test login below interfaces

Test login BMC debug console (UART)

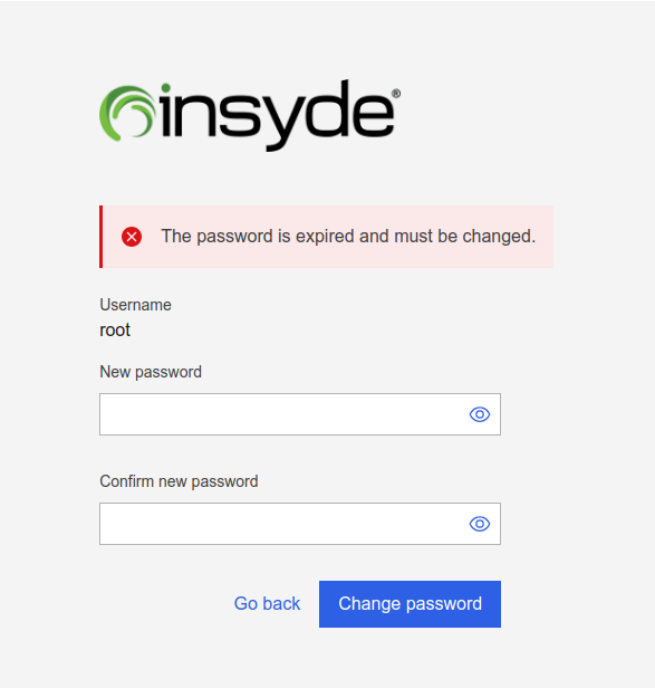
The user will be requested to update password before login

```
Phosphor OpenBMC (Phosphor OpenBMC Project Reference Distro) nodistro.0 bmc-maca4bf018d03a8 ttyS4

bmc-maca4bf018d03a8 login: root
Password:
You are required to change your password immediately (administrator enforced).
Changing password for root.
Current password:
New password:
Retype new password:
Last login: Fri Aug 30 04:35:16 UTC 2024 from ::ffff:192.168.10.140 on pts/0
```

Test login WebUI

The user will be requested to update password before login



Test login with SSH

(Nvidia platform) The user will be requested to update password before login

```
roykuo@ROY-KUO:~$ ssh admin@192.168.88.230
admin@192.168.88.230's password:

You are required to change your password immediately (administrator enforced).
Password has expired. Please change it now.
Enter admin@192.168.88.230's old password:
Enter admin@192.168.88.230's new password:
Retype admin@192.168.88.230's new password:
admin@nvidia-obmc:~#
```

(Other platforms) Login will be denied until user to update password

```
roykuo@ROY-KUO:~$ ssh root@192.168.88.230
root@192.168.88.230's password:

You are required to change your password immediately (administrator enforced).
Permission denied, please try again.
root@192.168.88.230's password:

You are required to change your password immediately (administrator enforced).
Permission denied, please try again.
root@192.168.88.230's password:

You are required to change your password immediately (administrator enforced).
root@192.168.88.230: Permission denied (publickey,password).
roykuo@ROY-KUO:~$
```

Test IPMI command over LAN

IPMI command over LAN will be denied until user to update password

```
ipmitool -U admin -P 0penBmc -I lanplus -C 17 -H 192.168.88.230 raw 6 1
Error: Unable to establish IPMI v2 / RMCP+ session
```

Note: IPMI command over system interface (KCS / SSIF / BT) will not be affected. User can update password via system interface before login.